

CHAPTER 6 Price Volatility.....	193
Introduction	193
Definitions.....	194
Price Returns/Price Change.....	194
Average Return.....	194
Volatility.....	196
Covariance	196
Correlation	197
Dispersion	197
Value-at-Risk	197
Implied Volatility	198
Beta	198
Market Observations—Empirical Findings.....	199
Forecasting Stock Volatility	202
Volatility Models.....	202
HMA-VIX Adjustment Model.....	206
Determining Parameters via Maximum Likelihood Estimation.....	208
Measuring Model Performance	209
Root Mean Square Error (RMSE).....	210
Root Mean Z-Score Squared Error (RMZSE)	210
Outlier Analysis.....	211
Results.....	211
Problems Resulting from Relying on Historical Market Data for Covariance Calculations	214
Factor Models	221
Types of Factor Models.....	227
 CHAPTER 7 Advanced Algorithmic Forecasting Techniques.....	 235
Introduction	235
Trading Cost Equations	235
Model Inputs.....	236
Trading Strategy.....	237
Percentage of Volume	237
Trading Rate	238
Trade Schedule	238
Comparison of <i>POV</i> rate to Trade Rate.....	239
Trading Time	239
Trading Risk Components	240